

Exercise-01: Ranking and Window function

Code:

-- Step 1: Create Database

```
CREATE DATABASE cts;
```

```
USE cts;
```

-- Step 2: Create Products Table

```
CREATE TABLE Products1
```

```
(
```

```
    ProductID INT PRIMARY KEY,
```

```
    ProductName VARCHAR(100),
```

```
    Category VARCHAR(50),
```

```
    Price DECIMAL(10,2)
```

```
);
```

-- Step 3: Insert Sample Data

```
INSERT INTO Products1(ProductID, ProductName, Category, Price)
```

```
VALUES
```

```
(1,'Laptop','Electronics',75000),
```

```
(2,'Mobile','Electronics',50000),
```

```
(3,'Tablet','Electronics',50000),
```

```
(4,'Headphones','Electronics',3000),
```

```
(5,'TV','Electronics',60000),
```

```
(6,'Sofa','Furniture',40000),
```

```
(7,'Chair','Furniture',5000),
```

```
(8,'Dining Table','Furniture',25000),
```

```
(9,'Bed','Furniture',40000),
```

```
(10,'Cupboard','Furniture',35000),
```

```
(11,'Shirt','Clothing',1500),
```

```
(12,'Jeans','Clothing',2500),
```

```
(13,'Jacket','Clothing',5000),
```

```
(14,'T-Shirt','Clothing',1500),
```

```
(15,'Sweater','Clothing',3000);
```

-- Display Table

```
SELECT * FROM Products1;
```

Output:

Result Grid   Filter Rows: Ex				
	ProductID	ProductName	Category	Price
▶	1	Laptop	Electronics	75000.00
	2	Mobile	Electronics	50000.00
	3	Tablet	Electronics	50000.00
	4	Headphones	Electronics	3000.00
	5	TV	Electronics	60000.00
	6	Sofa	Furniture	40000.00
	7	Chair	Furniture	5000.00
	8	Dining Table	Furniture	25000.00
	9	Bed	Furniture	40000.00
	10	Cupboard	Furniture	35000.00
	11	Shirt	Clothing	1500.00
	12	Jeans	Clothing	2500.00
	13	Jacket	Clothing	5000.00
	14	T-Shirt	Clothing	1500.00
	15	Sweater	Clothing	3000.00
•	NULL	NULL	NULL	NULL

-- ROW_NUMBER()

-- Top 3 expensive products in each category

```
SELECT *
FROM
(
    SELECT
        ProductID,
        ProductName,
        Category,
        Price,
        ROW_NUMBER() OVER
        (
            PARTITION BY Category
            ORDER BY Price DESC
        ) AS rn
```

```

FROM Products1
) t
WHERE rn <= 3;
Output:

```

Result Grid  Filter Rows: Export: 					
	ProductID	ProductName	Category	Price	rn
▶	13	Jacket	Clothing	5000.00	1
	15	Sweater	Clothing	3000.00	2
	12	Jeans	Clothing	2500.00	3
	1	Laptop	Electronics	75000.00	1
	5	TV	Electronics	60000.00	2
	2	Mobile	Electronics	50000.00	3
	6	Sofa	Furniture	40000.00	1
	9	Bed	Furniture	40000.00	2
	10	Cupboard	Furniture	35000.00	3

-- RANK()

```

SELECT *
FROM
(
    SELECT
        ProductID,
        ProductName,
        Category,
        Price,
        RANK() OVER
        (
            PARTITION BY Category
            ORDER BY Price DESC
        ) AS rnk
    FROM Products1
) t
WHERE rnk <= 3;
Output:

```

Result Grid  Filter Rows: Export: 					
	ProductID	ProductName	Category	Price	rnk
▶	13	Jacket	Clothing	5000.00	1
	15	Sweater	Clothing	3000.00	2
	12	Jeans	Clothing	2500.00	3
	1	Laptop	Electronics	75000.00	1
	5	TV	Electronics	60000.00	2
	2	Mobile	Electronics	50000.00	3
	3	Tablet	Electronics	50000.00	3
	6	Sofa	Furniture	40000.00	1
	9	Bed	Furniture	40000.00	1
	10	Cupboard	Furniture	35000.00	3

-- DENSE_RANK()

```

SELECT *
FROM
(
    SELECT
        ProductID,
        ProductName,
        Category,
        Price,
        DENSE_RANK() OVER
        (
            PARTITION BY Category
            ORDER BY Price DESC
        ) AS drnk
    FROM Products1
) t
WHERE drnk <= 3;

```

Output:

Result Grid					
Filter Rows:					
Export: 					
	ProductID	ProductName	Category	Price	rnk
▶	13	Jacket	Clothing	5000.00	1
	15	Sweater	Clothing	3000.00	2
	12	Jeans	Clothing	2500.00	3
	1	Laptop	Electronics	75000.00	1
	5	TV	Electronics	60000.00	2
	2	Mobile	Electronics	50000.00	3
	3	Tablet	Electronics	50000.00	3
	6	Sofa	Furniture	40000.00	1
	9	Bed	Furniture	40000.00	1
	10	Cupboard	Furniture	35000.00	2
	8	Dining Table	Furniture	25000.00	3

-- Compare All Ranking Functions

SELECT

ProductID,
 ProductName,
 Category,
 Price,

ROW_NUMBER() OVER
 (
 PARTITION BY Category
 ORDER BY Price DESC
) AS rn,

RANK() OVER
 (
 PARTITION BY Category
 ORDER BY Price DESC
) AS rnk,

DENSE_RANK() OVER

```
(
  PARTITION BY Category
  ORDER BY Price DESC
) AS drnk
```

```
FROM Products1
ORDER BY Category, Price DESC;
```

Output:

Result Grid							
		Filter Rows:		Export:		Wrap Cell Conte	
	ProductID	ProductName	Category	Price	rn	rnk	drnk
▶	13	Jacket	Clothing	5000.00	1	1	1
	15	Sweater	Clothing	3000.00	2	2	2
	12	Jeans	Clothing	2500.00	3	3	3
	11	Shirt	Clothing	1500.00	4	4	4
	14	T-Shirt	Clothing	1500.00	5	4	4
	1	Laptop	Electronics	75000.00	1	1	1
	5	TV	Electronics	60000.00	2	2	2
	2	Mobile	Electronics	50000.00	3	3	3
	3	Tablet	Electronics	50000.00	4	3	3
	4	Headphones	Electronics	3000.00	5	5	4
	6	Sofa	Furniture	40000.00	1	1	1
	9	Bed	Furniture	40000.00	2	1	1
	10	Cupboard	Furniture	35000.00	3	3	2
	8	Dining Table	Furniture	25000.00	4	4	3
	7	Chair	Furniture	5000.00	5	5	4